

GraphML-FS

Feature synthesis as the atomic search unit — controller /
worker split, typed operator bank

Recap

Prior deck: zhuconv.github.io/slides-hub/graphagent-4023

Where we left off — GraphLoomer

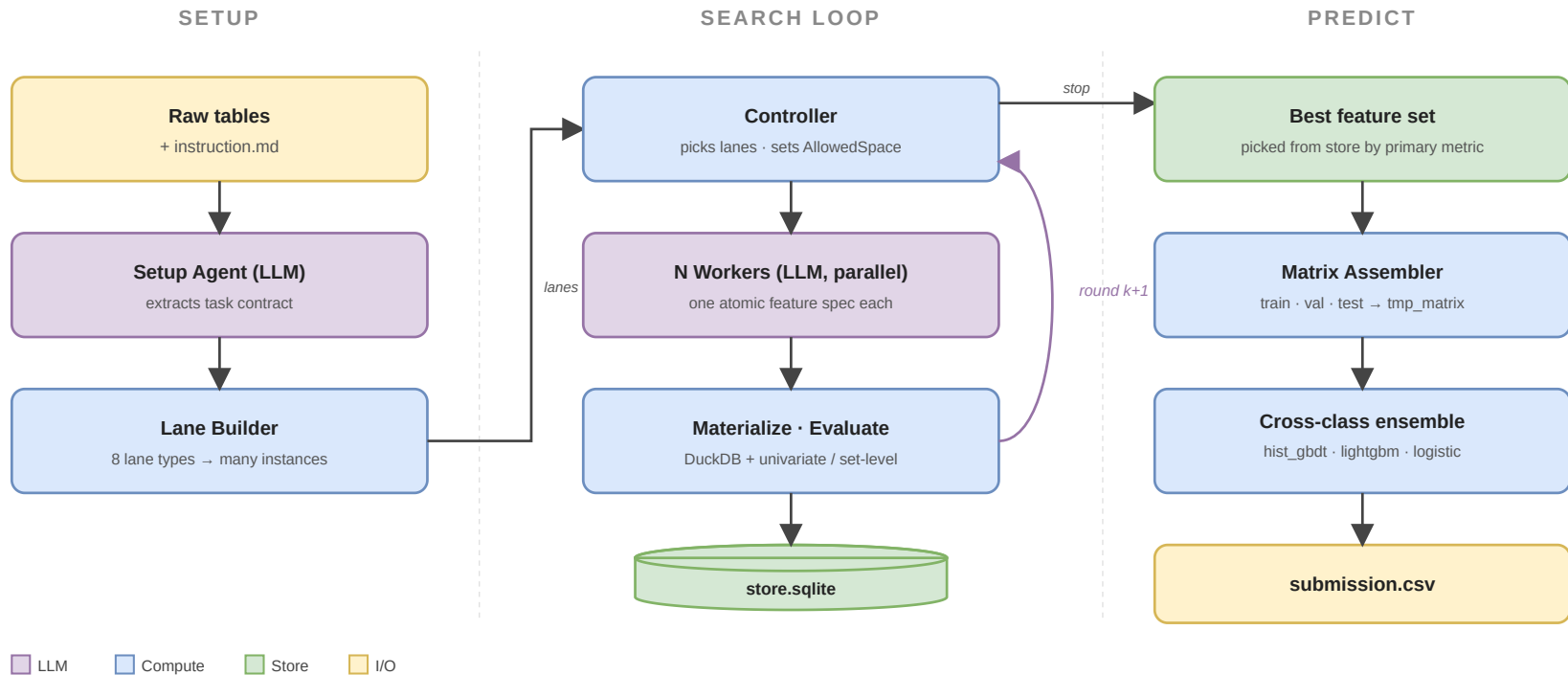
- Bilevel pipeline: outer model type, inner feature ops from a catalog
- Whole `solution.py` was the unit of search; one LLM agent looped Plan → Code → Execute → Refine
- 3rd of 4 end-to-end on GraphTestbed; failure-pattern features transferred to MLEvolve

What was missing

- No parallelism inside a round — one agent, one solution at a time
- Errors hard to localise — a feature bug crashed the whole pipeline
- Free-form Python outran the validator → debug churn dominated wall time

This deck: rebuild the harness around three architectural decisions, each fixing one of the gaps above.

At a glance — three stages, one loop



Setup runs once. Search loops until the controller stops. Predict materialises the best feature set into a single submission. Three pillars on the next three slides.

Pillar 1 — Controller $\perp$ Worker, parallel rounds

What

- **Controller** — rule-based, no LLM. Enumerates lanes, sets `AllowedSpace`, decides stop.
- **Worker** — LLM. Stateless. Emits one `AtomicFeatureProposalBatch` for the lane it's given.
- **N workers per round, in parallel.**

Why it matters

- Round wall-time bounded by the slowest worker, not by total prompt count
- Errors localise to a single lane / spec
- Borrowed: AI-Build-AI's role separation; MLEvolve's per-node LLM expansion

GraphLoomer used one LLM looped on a whole pipeline — sequential, long context, blame hard to attribute. Splitting roles is the single biggest source of speed-up.

Pillar 2 — atomic feature spec is the search unit

What

A single search step produces:

```
spec = (lane, target_entity, path[],  
        value_col, agg, window)
```

Canonicalised + hashed → deduped across rounds → persisted
in `store.sqlite`.

Why it matters

- Materialize + univariate eval is **per spec** → embarrassingly parallel
- Round $k+1$ reuses rounds $0\dots k$'s matrices
- LLM mistakes scope to one feature, not the whole pipeline
- "What paid off" → next round's `AllowedSpace`

The downstream — impute → fit `hist_gbd`t / `lightgbm` / logistic → threshold sweep on val → CSV — is fixed code. The LLM never rewrites it.

Pillar 3 — typed operator bank: 8 lane *types*

Local lane types

- `direct_numeric` · pass-through
- `local_column_transform` · log1p / clip / bucket
- `local_expr` · DSL expr over base cols
- `sparse_cross` · cat × cat hashed cross

Relational lane types (time-respecting)

- `shared_key_group_aggregate` · join + group-by
- `account_history` · windowed history aggs
- `account_static_aggregate` · snapshot rollup
- `account_mixed_history` · 3-step relational walk

Why a closed, typed set

- Each type fixes input dtype, output dtype, time policy, allowed aggs/windows
- Validator rejects illegal specs **before** SQL runs
- LLM degrees of freedom capped to options the validator understands
- A new lane type is a code change, not an LLM choice

The Worker LLM picks SQL *within* a lane instance — never invents a new lane type. Sonnet still hits 37 % materialize-fail with this guardrail; free-form would be much worse.

Results — first on 3 / 4 GraphTestbed tasks

`graphfs-claude-sonnet-4-6` on the public leaderboard ([lanczos/graphtestbed](https://lanczos.github.io/graph-testbed/)):

Task	Metric	GraphML-FS	rank	next-best
<code>figraph</code>	AUC-ROC	0.895	#1	0.890 (open-aibuildai)
<code>arxiv-citation</code>	AUC-ROC	0.789	#1	0.777 (open-aibuildai)
<code>ibm-aml</code>	F1 (minority)	0.184	#1	0.171 (open-aibuildai)
<code>ieee-fraud-detection</code>	AUC-ROC	0.921	#3	0.928 (aibuildai)

Same `claude-sonnet-4-6` as the AI-Build-AI runs — the harness is the differentiator, not the model.

Bigger picture — graph learning is operator learning

Where graph FMs hard-code the operator

- GNN / OFA — fixed local message-passing
- Tokenizer Transformers (Graphormer, OpenGraph) — PE / SVD / token bias
- Hybrid GT (GraphGPS) — local MPNN + global attention
- PFN / G2T-FM — graph adapter into a tabular FM

Pivot — operator scaling

First-class searchable modules:

- Construction — A , kNN, latent
- Reduction — sum / mean / attn
- Multi-hop — A^2 , PPR, RW, heat
- Structure — LapPE, RWSE, motif
- Head — MLP / RF / XGB / PFN

NN's job: select, compose, calibrate, fuse.

GraphML-FS's 8 lane *types* are this thesis instantiated on the tabular-graph side. Generalising to end-to-end GraphFM training is the longer arc — operator scaling vs architecture scaling.

Next steps

1 — Three-axis bench

Per system × task × LLM: **performance**, **API cost**, **wall-time**. Compare GraphML-FS / MLEvolve / open-aibuildai across `sonnet-4-6` / `opus-4-7` / `gpt-5.4` .
Hypothesis: typed bank caps tokens → lowest cost-per-point.

2 — Close the autoresearch ceiling

Task	now	ceiling	Δ
<code>arxiv-citation</code>	0.789	0.824	-0.035
<code>figraph</code>	0.895	0.940	-0.045
<code>ibm-aml</code>	0.184	0.591	-0.407

K-fold OOF · calibration · auto-NS · lane meta-prior.

3 — Ship as skills + tools

Drop-ins for MLEvolve / open-aibuildai: `graphfs.search` (matrix + lift) and `graphfs.lane_eval` (score one spec on val). Distribute as MCP server — zero host-controller change.

4 — Reminder · pre-NN vs post-NN aggregation

Current 8 lanes are **pre-NN** (column → downstream). Post-NN = wrap a model whose preds aggregate. Forces a differentiable / non-differentiable lane split. **Parked**.